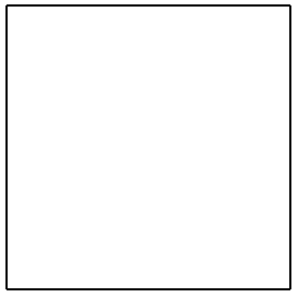
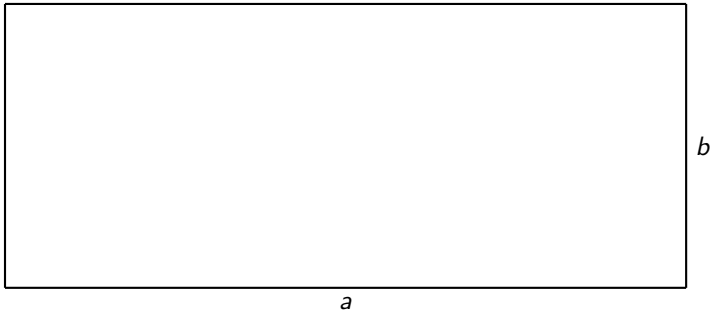
Riemann Integration

Riemann Integral: Motivation



The area of a unit square is 1 square unit.

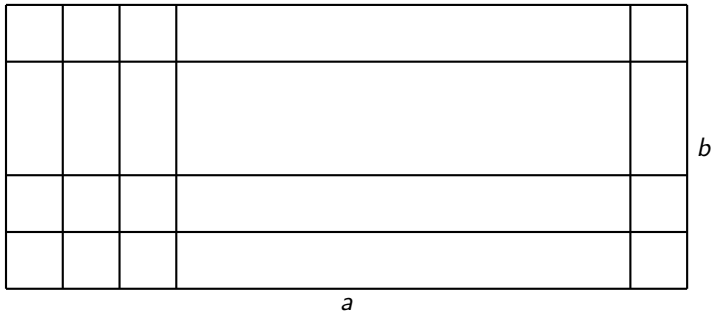
The Area of a Rectangle



Divide into unit squares? Yes!

The Area of a Rectangle

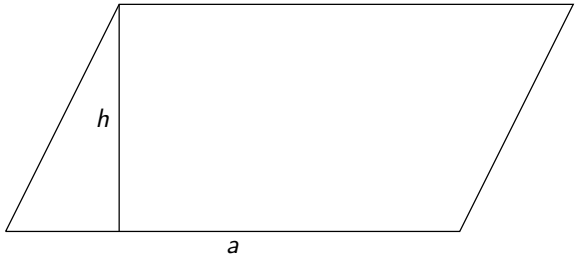
Divide into unit squares and count them!



Area of the rectangle is ab square units.

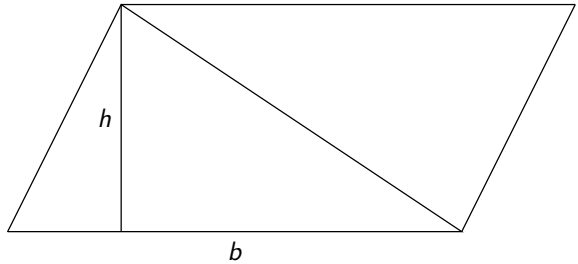
Note: The formula is valid for all non-negative real numbers.

The Area of a Parallelogram



The area is ah square units.

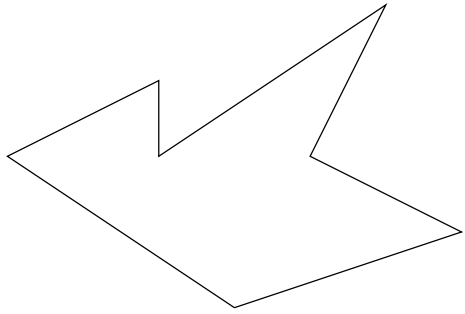
The Area of a Triangle



Area of the parallelogram is bh square units.

Hence the area of the triangle is $\frac{1}{2}bh$ square units.

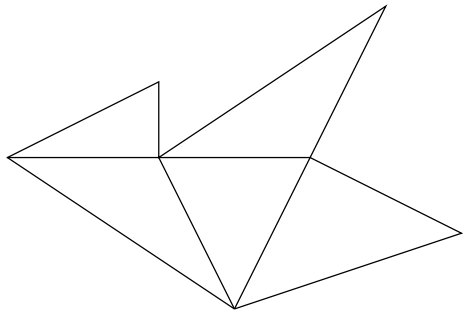
The Area of a Polygonal Region



Triangulate!

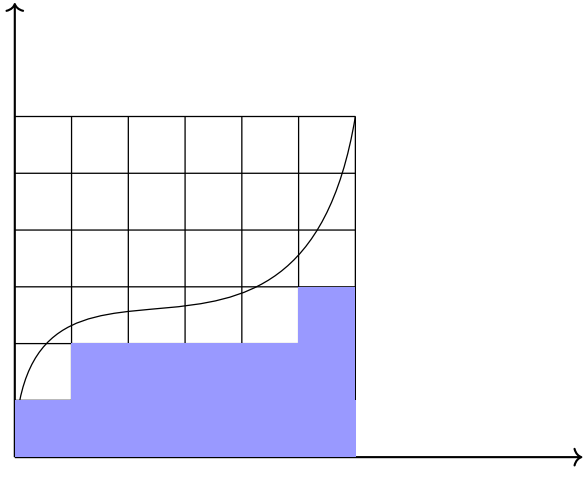
The Area of a Polygonal Region

Triangulate and add the areas of the triangles!

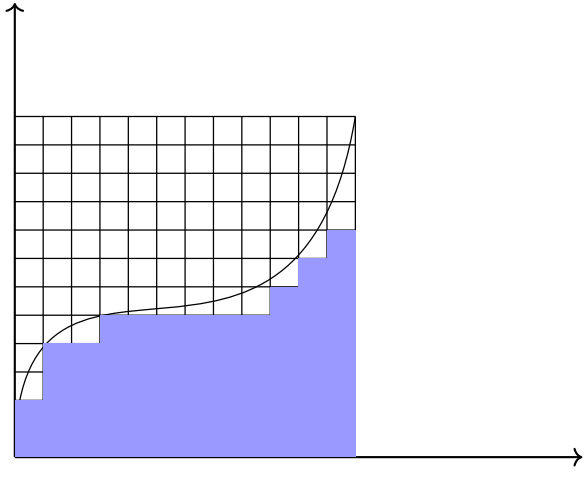


The area of the polygon is the sum of the areas of the triangles!

Computing area of an irregular object



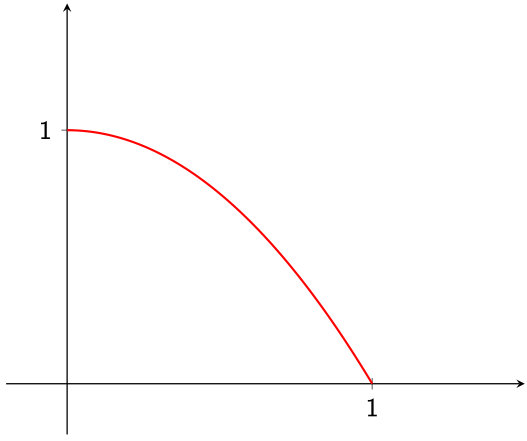
Increasing the Number of Cuts



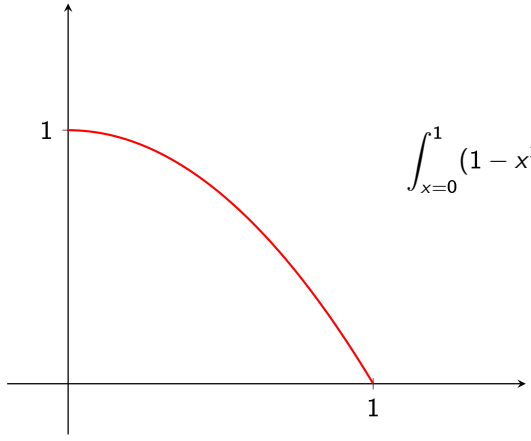
Note

1. Thus increase in number of cuts leads to a better estimate of the required area.
2. How to find the exact area? By cutting into infinitely many small square? Does it make sense? It does! Riemann Integration does exactly that!

Area Under the Curve $y = 1 - x^2$ in the First Quadrant

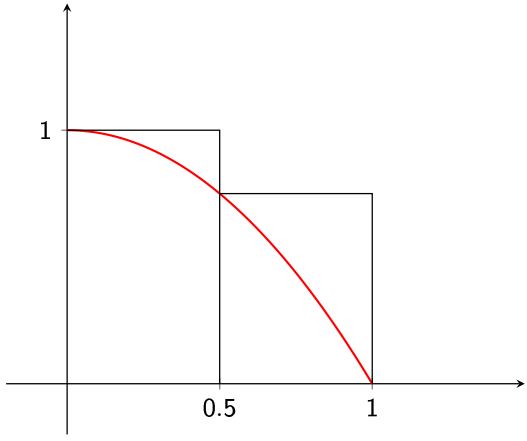


Area Under the Curve $y = 1 - x^2$ in the First Quadrant



$$\int_{x=0}^{1} (1 - x^2) dx = \frac{2}{3} \approx 0.667$$

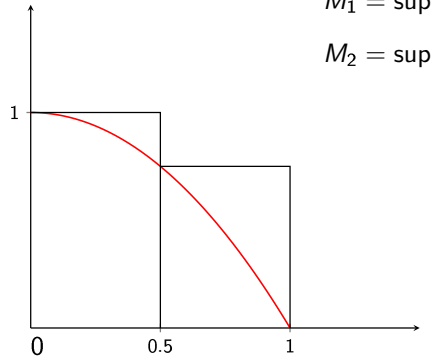
Area under the curve $y = 1 - x^2$: An Upper Estimate



An Upper Estimate

$$M_1 = \sup\{f(x) : x \in [0, 0.5]\} = 1$$

$$M_2 = \sup\{f(x) : x \in [0.5, 1]\} = \tfrac{3}{4}$$

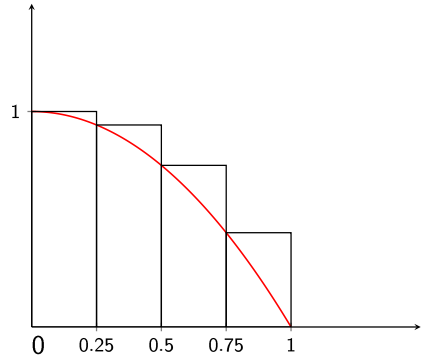


The total area of the two rectangles is

$$1 \cdot 0.5 + \frac{3}{4} \cdot 0.5 = 0.875.$$

This is an upper bound on the required area. It is called an **upper sum**.

Area Under the Curve $y = 1 - x^2$: A Better Upper Estimate

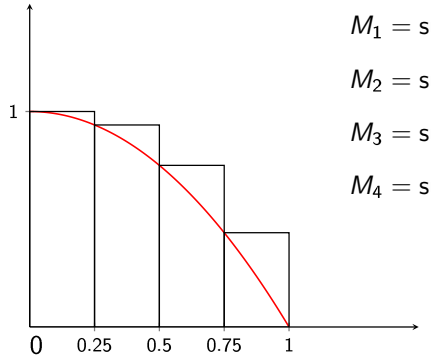


The total area of the four rectangles is

$$1 \cdot 0.25 + \frac{15}{16} \cdot 0.25 + \frac{3}{4} \cdot 0.25 + \frac{7}{16} \cdot 0.25 = 0.78125.$$

This is a better upper bound on the required area. Moreover, it is an **upper sum.**

Area Under the Curve $y = 1 - x^2$: A Better Upper Estimate



$$M_1 = \sup\left\{1 - x^2 : x \in [0, 0.25]\right\} = 1$$

$$M_2 = \sup\left\{1 - x^2 : x \in [0.25, 0.5]\right\} = \frac{15}{16}$$

$$M_3 = \sup\left\{1 - x^2 : x \in [0.5, 0.75]\right\} = \frac{3}{4}$$

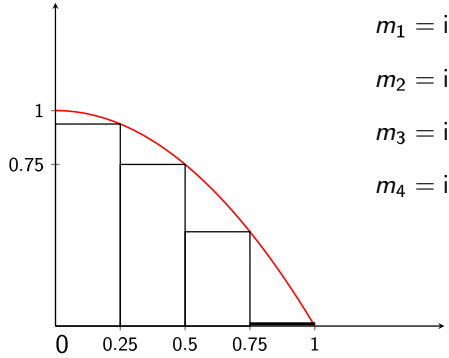
$$M_4 = \sup\left\{1 - x^2 : x \in [0.75, 1]\right\} = \frac{7}{16}$$

The total area of the four rectangles is

$$1 \cdot 0.25 + \frac{15}{16} \cdot 0.25 + \frac{3}{4} \cdot 0.25 + \frac{7}{16} \cdot 0.25 = 0.78125.$$

This is a better upper bound on the required area. Moreover, it is an **upper sum.**

Area Under the Curve $y = 1 - x^2$: A Lower Estimate



$$m_1 = \inf\left\{1 - x^2 : x \in [0, 0.25]\right\} = \tfrac{15}{16}$$

$$m_2 = \inf\left\{1 - x^2 : x \in [0.25, 0.5]\right\} = \tfrac{3}{4}$$

$$m_3 = \inf\left\{1 - x^2 : x \in [0.5, 0.75]\right\} = \tfrac{7}{16}$$

$$m_4 = \inf\left\{1 - x^2 : x \in [0.75, 1]\right\} = 0$$

The total area of the four rectangles is

$$\frac{15}{16} \cdot 0.25 + \frac{3}{4} \cdot 0.25 + \frac{7}{16} \cdot 0.25 + 0 \cdot 0.25 = 0.53125.$$

It is called a **lower sum**.

Definition (Partition)

Let $[a, b]$ be a closed interval. Then a set $P = \{x_0, x_1, x_2, \ldots, x_n\}$ is called a partition of the interval if

$$a = x_0 \leq x_1 \leq x_2 \leq \cdots \leq x_n = b.$$

Notation

If $P = \{x_0, x_1, x_2, \ldots, x_n\}$ is a partition of an interval $[a, b]$, we denote the length of the ith subinterval $[x_{i-1}, x_i]$ by

$$\Delta x_i = x_i - x_{i-1}, \quad i = 1, 2, \ldots, n.$$

Note

$$\sum_{i=1}^{n} \Delta x_i = (x_1 - x_0) + (x_2 - x_1) + \cdots + (x_n - x_{n-1}) = x_n - x_0 = b - a.$$

Examples

Consider the interval $[0, 1]$. The following are some partitions of it:

1. $P_1 = \{0, 0.5, 1\}$.
2. $P_2 = \{0, 0.25, 0.5, 1\}$.
3. $P_3 = \{0, 0.25, 0.5, 0.75, 1\}$.
4. $P_4 = \{0, 0.1, 0.3, 0.7, 1\}$.
5. $P_5 = \{0, 0.2, 0.4, 0.6, 0.8, 1\}$.

For partition P_1,

$$x_0 = 0, x_1 = 0.5, x_2 = 1 \quad \text{and} \quad \Delta x_1 = 0.5, \Delta x_2 = 0.5.$$

For partition P_2,

$$x_0 = 0, x_1 = 0.25, x_2 = 0.5, x_3 = 1 \quad \text{and} \quad \Delta x_1 = 0.25, \Delta x_2 = 0.25, \Delta x_3 = 0.5.$$

Definition (Upper Riemann Sum, Lower Riemann Sum)

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be a partition of $[a, b]$.

For each $i = 1, 2, \ldots, n$, let

$$M_i = \sup\{f(x) \mid x \in [x_{i-1}, x_i]\} \quad \text{and} \quad m_i = \inf\{f(x) \mid x \in [x_{i-1}, x_i]\}.$$

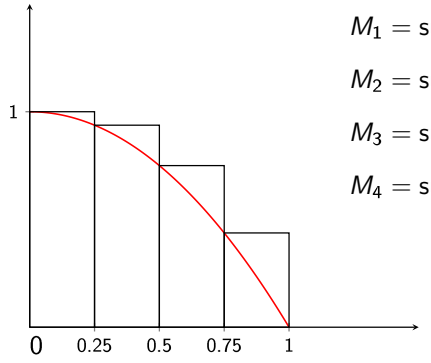
Then the **upper Riemann sum** corresponding to the partition P is

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i.$$

Similarly, the **lower Riemann sum** corresponding to P is

$$L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i.$$

Example



$$M_1 = \sup\left\{1 - x^2 : x \in [0, 0.25]\right\} = 1$$

$$M_2 = \sup\left\{1 - x^2 : x \in [0.25, 0.5]\right\} = \frac{15}{16}$$

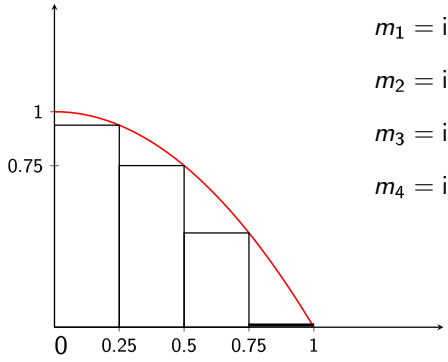
$$M_3 = \sup\left\{1 - x^2 : x \in [0.5, 0.75]\right\} = \frac{3}{4}$$

$$M_4 = \sup\left\{1 - x^2 : x \in [0.75, 1]\right\} = \frac{7}{16}$$

The upper Riemann sum corresponding to the partition $P = \{0, 0.25, 0.5, 0.75, 1\}$ is

$$U(P, f) = M_1\Delta x_1 + M_2\Delta x_2 + M_3\Delta x_3 + M_4\Delta x_4 = 1 \cdot 0.25 + \frac{15}{16} \cdot 0.25 + \frac{3}{4} \cdot 0.25 + \frac{7}{16} \cdot 0.25 = 0.78125.$$

Example



$$m_1 = \inf\left\{1 - x^2 : x \in [0, 0.25]\right\} = \tfrac{15}{16}$$

$$m_2 = \inf\left\{1 - x^2 : x \in [0.25, 0.5]\right\} = \tfrac{3}{4}$$

$$m_3 = \inf\left\{1 - x^2 : x \in [0.5, 0.75]\right\} = \tfrac{7}{16}$$

$$m_4 = \inf\left\{1 - x^2 : x \in [0.75, 1]\right\} = 0$$

The lower Riemann sum corresponding to the partition $P = \{0, 0.25, 0.5, 0.75, 1\}$ is

$$L(P, f) = m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 + m_4\Delta x_4 = \frac{15}{16}\cdot 0.25 + \frac{3}{4}\cdot 0.25 + \frac{7}{16}\cdot 0.25 + 0\cdot 0.25 = 0.53125.$$

Note

In the preceding example, $L(P, f) \leq U(P, f)$. It is always true. In fact, the following theorem is true:

Lemma

Let $f : [a, b] \to \mathbb{R}$ be a bounded function such that $m \leq f(x) \leq M$. Then for any partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$ of $[a, b]$

$$m(b - a) \leq L(P, f) \leq U(P, f) \leq M(b - a).$$

Proof:

Let

$$M_i = \sup\{f(x) \mid x \in [x_{i-1}, x_i]\} \quad \text{and} \quad m_i = \inf\{f(x) \mid x \in [x_{i-1}, x_i]\} \quad \text{for } i = 1, 2, \ldots, n.$$

Then

$$\begin{aligned}
m \le m_i \le M_i \le M \quad &\Rightarrow \quad m\Delta x_i \le m_i \Delta x_i \le M_i \Delta x_i \le M \Delta x_i \\
&\Rightarrow \quad \sum_{i=1}^{n} m\Delta x_i \le \sum_{i=1}^{n} m_i \Delta x_i \le \sum_{i=1}^{n} M_i \Delta x_i \le \sum_{i=1}^{n} M \Delta x_i \\
&\Rightarrow \quad m \sum_{i=1}^{n} \Delta x_i \le \sum_{i=1}^{n} m_i \Delta x_i \le \sum_{i=1}^{n} M_i \Delta x_i \le M \sum_{i=1}^{n} \Delta x_i \\
&\Rightarrow \quad m(b-a) \le L(P, f) \le U(P, f) \le M(b-a).
\end{aligned}$$

Corollary

Let f be a bounded real-valued function on $[a, b]$. Then the set of all lower sums is bounded above. And the set of all upper sums is bounded below.

Refinement of a Partition and Common Refinement

Definition (Refinement, Common Refinement)

1. Let P_1 be a partition of $[a, b]$. Then a partition P_2 of $[a, b]$ is called a **refinement** of P_1 if $P_1 \subseteq P_2$.
2. Let P_1 and P_2 be partitions of $[a, b]$. Then $P_1 \cup P_2$ is called a **common refinement** of P_1 and P_2.

Examples:

1. Consider the interval $[0, 1]$ and its partitions $P_1 = \{0, 0.5, 1\}$ and $P_2 = \{0, 0.5, 0.75, 1\}$. Here $P_1 \subseteq P_2$. So, P_2 is a refinement of P_1.
2. Consider the interval $[0, 1]$ and its partitions $P_1 = \{0, 0.25, 0.5, 1\}$ and $P_2 = \{0, 0.5, 0.75, 1\}$. Then their common refinement is $P_1 \cup P_2 = \{0, 0.25, 0.5, 0.75, 1\}$.

Lemma

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let P_1 and P_2 be partitions of $[a, b]$ such that P_2 is a refinement of P_1. Then

$$L(P_1, f) \leq L(P_2, f) \leq U(P_2, f) \leq U(P_1, f).$$

Theorem

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let P_1 and P_2 be any partitions of $[a, b]$. Then

$$L(P_1, f) \leq U(P_2, f).$$

Proof: Let $Q = P_1 \cup P_2$ be the common refinement of P_1 and P_2. Then by the above lemma

$$L(P_1, f) \leq L(Q, f) \leq U(Q, f) \leq U(P_2, f).$$

Definition

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. The **upper Riemann integral** of f over $[a, b]$ is

$$\overline{\int_a^b} f(x)dx = \inf \{U(P, f) : \ P \text{ is a partition of } [a, b]\} = \inf U(P, f).$$

The **lower Riemann integral** of f over $[a, b]$ is

$$\underline{\int_a^b} f(x)dx = \sup \{L(P, f) : \ P \text{ is a partition of } [a, b]\} = \sup L(P, f).$$

Note: From the theorem on the preceding slide, we have

$$\underline{\int_a^b} f(x)dx \leq \overline{\int_a^b} f(x)dx.$$

Definition

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. We say f **is Riemann integrable on** $[a, b]$ if the upper and lower Riemann integrals are equal, that is,

$$\overline{\int_a^b} f(x)dx = \underline{\int_a^b} f(x)dx.$$

In this case, we say f is Riemann integrable on $[a, b]$ (or $f \in \mathcal{R}$) and denote the common value by

$$\int_a^b f(x)dx = \overline{\int_a^b} f(x)dx = \underline{\int_a^b} f(x)dx.$$

Example

Show that $f(x) = k$ (a constant function) on $[a, b]$ is Riemann integrable.

Solution: Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be any partition of $[a, b]$. Then

$$M_i = \sup\{f(x) : x_{i-1} \le x \le x_i\} = \sup\{k : x_{i-1} \le x \le x_i\} = k.$$

$$m_i = \inf\{f(x) : x_{i-1} \le x \le x_i\} = \inf\{k : x_{i-1} \le x \le x_i\} = k.$$

Thus the upper Riemann sum corresponding to partition P is

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i = \sum_{i=1}^{n} k \Delta x_i = k \sum_{i=1}^{n} \Delta x_i = k(b - a),$$

and the lower Riemann sum corresponding to partition P is

$$L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i = \sum_{i=1}^{n} k \Delta x_i = k \sum_{i=1}^{n} \Delta x_i = k(b - a).$$

Note that we computed $U(P, f)$ and $L(P, f)$ for an arbitrary partition P on $[a, b]$.

Thus the upper Riemann integral of f over $[a, b]$ is

$$\overline{\int_a^b} f(x)dx = \inf\{U(P, f) : P \text{ is a partition of } [a, b]\} = \inf\{k(b - a)\} = k(b - a).$$

And the lower Riemann integral of f over $[a, b]$ is

$$\underline{\int_a^b} f(x)dx = \sup\{L(P, f) : P \text{ is a partition of } [a, b]\} = \sup\{k(b - a)\} = k(b - a).$$

So, the upper and lower Riemann integrals are equal. Hence the function is Riemann integrable and

$$\int_a^b f(x)dx = \int_a^b kdx = k(b - a).$$

Example of a non-Riemann Integrable Bounded Function

Let $f : [0,1] \longrightarrow R$ be the function defined by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is a rational number} \\ 0, & \text{if } x \text{ is an irrational number.} \end{cases}$$

Show that $f \notin \mathcal{R}$ (f is not a Riemann integrable function).

Solution: Note that $0 \leq f(x) \leq 1$ for all $x \in [0,1]$. So, it is a bounded function on $[0,1]$.

Let $P = \{0 = x_0, x_1, \ldots, x_n = 1\}$ be any partition of $[0,1]$. That is

$$0 = x_0 \leq x_1 \leq \ldots \leq x_n = 1.$$

Then

$$M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\} = \sup\{1, 0\} = 1$$

and

$$m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\} = \inf\{1, 0\} = 0.$$

Thus, the upper Riemann sum of f corresponding to the partition P is

$$U(P,f) = \sum_{i=1}^{n} M_i \Delta x_i = \sum_{i=1}^{n} \Delta x_i = 1 - 0 = 1$$

and the lower Riemann sum of f corresponding to the partition P is

$$L(P,f) = \sum_{i=1}^{n} m_i \Delta x_i = 0.$$

Note that we computed $U(P,f)$ and $L(P,f)$ for an arbitrary partition P on $[0,1]$.

Thus the upper Riemann integral of f over $[0,1]$ is

$$\overline{\int_0^1} f(x)dx = \inf\{U(P,f) \ : \ P \text{ is a partition of } [0,1]\} = \inf\{1\} = 1,$$

and the lower Riemann integral of f over $[0,1]$ is

$$\underline{\int_0^1} f(x)dx = \sup\{L(P,f) \ : \ P \text{ is a partition of } [a,b]\} = \sup\{0\} = 0.$$

Thus

$$\overline{\int_0^1} f(x)dx \neq \underline{\int_0^1} f(x)dx.$$

Hence f is not Riemann integrable on $[0,1]$.

Homework

Let $f : [0, 1] \to R$ be a function defined as

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is a rational number} \\ -1, & \text{if } x \text{ is an irrational number.} \end{cases}$$

Prove or disprove the following: (i) f is a Riemann integrable function and (ii) $|f|$ is a Riemann integrable function.



Theorem

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Then f is Riemann integrable if and only if for every $\epsilon > 0$, there exists a partition P such that

$$U(P, f) - L(P, f) < \epsilon.$$

Example

Show that the function $f(x) = x$ is Riemann integrable. Also find $\int_a^b f(x)dx$.

Solution: Let $h = \frac{b-a}{n}$ and $P_n = \{a = x_0, x_1 = a + h, x_2 = a + 2h, \ldots, x_n = a + nh = b\}$. Then, we have

$$m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\} = \inf\{x : a + (i-1)h \leq x \leq a + ih\} = a + (i-1)h.$$

Thus

$$\begin{aligned}
L(P_n, f) &= \sum_{i=1}^{n} m_i \Delta x_i \\
&= \sum_{i=1}^{n} (a + (i-1)h)\, h \\
&= ah\sum_{i=1}^{n}(1) + h^2\sum_{i=1}^{n}(i-1) \\
&= ahn + h^2\,(0 + 1 + 2 + \ldots + (n-1)) \\
&= a \times nh + h^2\,\tfrac{n(n-1)}{2} \\
&= a(b-a) + \tfrac{(b-a)^2}{n^2}\,\tfrac{n(n-1)}{2} \\
&= a(b-a) + \tfrac{(b-a)^2}{2}(1 - \tfrac{1}{n}).
\end{aligned}$$

Similarly,

$$M_i = \sup\{f(x) : x_{i-1} \le x \le x_i\} = \sup\{x : a + (i-1)h \le x \le a + ih\} = a + ih,$$

and

$$\begin{aligned}
U(P_n, f) &= \sum_{i=1}^{n} M_i \Delta x_i \\
&= \sum_{i=1}^{n} (a + ih)\, h \\
&= ah\sum_{i=1}^{n}(1) + h^2\sum_{i=1}^{n}(i) \\
&= ahn + h^2\,(1 + 2 + \ldots + n) \\
&= a \times nh + h^2\,\tfrac{n(n+1)}{2} \\
&= a(b-a) + \tfrac{(b-a)^2}{n^2}\,\tfrac{n(n+1)}{2} \\
&= a(b-a) + \tfrac{(b-a)^2}{2}(1 + \tfrac{1}{n}).
\end{aligned}$$

We observe that

$$\lim_{n \longrightarrow \infty} L(P_n, f) = a(b-a) + \frac{(b-a)^2}{2}(1-0) = \frac{b^2 - a^2}{2}$$

and

$$\lim_{n \longrightarrow \infty} U(P_n, f) = a(b-a) + \frac{(b-a)^2}{2}(1+0) = \frac{b^2 - a^2}{2}.$$

As

$$\lim_{n \longrightarrow \infty} U(P_n, f) = \lim_{n \longrightarrow \infty} L(P_n, f),$$

we have

$$\lim_{n \longrightarrow \infty} [U(P_n, f) - L(P_n, f)] = 0.$$

Let $\epsilon > 0$ be an arbitrary positive number. From the definition of convergence of sequences, we can find a positive integer N such that for any $n \geq N$,

$$U(P_n, f) - L(P_n, f) < \epsilon.$$

So, it follows that corresponding to the $\epsilon > 0$ given, there is a partition P_n satisfying above inequality. Hence, by the previous theorem, the function is Riemann integrable.

Furthermore, as

$$L = L(P_n, f) \leq \int_a^b f(x)dx \leq U(P_n, f), = L$$

we have

$$\int_a^b f(x)dx = \lim_{n \longrightarrow \infty} U(P_n, f) = \frac{b^2 - a^2}{2}.$$

Homework

1. Let $f(x) = x^2$ be defined on $[a, b]$. Let $h = \frac{b-a}{n}$. Let $P_n = \{a = x_0, x_1 = a + h, x_2 = a + 2h, \ldots, x_n = a + nh = b\}$.
   (i) Compute $L(P_n, f)$ and $U(P_n, f)$.
   (ii) Find the limits of these lower and upper Riemann sums and conclude that the function is Riemann itegrable.
   (iii) Also, find the Riemann integral of $f(x)$.

2. Let $g(x) = x^2$ be defined on $[a, b]$ where $0 < a < b$. Let $h = \left(\frac{b}{a}\right)^{\frac{1}{n}}$. Let $Q_n = \left\{a = x_0, x_1 = ah, x_2 = ah^2, \ldots, x_n = ah^n = b\right\}$.
   (i) Compute $L(Q_n, g)$ and $U(Q_n, g)$.
   (ii) Find the limits of these lower and upper Riemann sums and conclude that the function is Riemann itegrable.
   (iii) Also, find the Riemann integral of $f(x)$.

3. Let $f(x) = 1 - x^2$ be defined on $[0, 1]$. Let $P_n = \left\{x_0 = 0, x_1 = \frac{1}{n}, x_2 = \frac{2}{n}, \ldots, x_n = \frac{n}{n} = 1\right\}$.
   (i) Compute $L(P_n, f)$ and $U(P_n, f)$.
   (ii) Argue that the function is Riemann integrable and find the Riemann integral.

Definition (Riemann sum)

Let $f(x)$ be a bounded real valued function defined on $[a, b]$. Let $P = \{x_0 = a, x_1, x_2, \ldots, x_n = b\}$ be a partition of $[a, b]$. Let $c_i \in [x_{i-1}, x_i]$, $1 \leq i \leq n$. Then

$$S_P = \sum_{i=1}^{n} f(c_i)\Delta x_i$$

is called a **Riemann sum** for f corresponding to the partition P.

Note:

$$L(P, f) \leq S_P \leq U(P, f).$$

Theorem (Riemann Integrability of Continuous Functions)

If a function f is continuous on the interval $[a, b]$, then it is Riemann integrable.

Moreover, if $h = \frac{b-a}{n}$ and $P_n = \{a = x_0, x_1 = a + h, x_2 = a + 2h, \ldots, x_n = a + nh = b\}$ is a partition of $[a, b]$ into equal subintervals, then

$$\lim_{n \longrightarrow \infty} L(P_n, f) = \lim_{n \longrightarrow \infty} U(P_n, f) = \int_a^b f(x)dx.$$

Hence if S_{P_n} is any Riemann sum corresponding to P_n, then

$$\lim_{n \longrightarrow \infty} S_{P_n} = \int_a^b f(x)dx.$$

Example

Find the Riemann integrable of the function $f(x) = x + x^2$ defined on $[0, 1]$.

Solution: Let us divide the interval $[0, 1]$ into n equal subintervals, each of length $\frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$. That is

$$P_n = \left\{ x_0 = 0, x_1 = \frac{1}{n}, \ldots, x_i = \frac{i}{n}, \ldots, x_n = 1 \right\}.$$

For each i, let $c_i = x_i = \frac{i}{n}$. Then $f(c_i) = c_i + c_i^2 = \frac{i}{n} + \frac{i^2}{n^2}$.

Then the Riemann sum corresponding to partition P_n is

$$\begin{aligned} S_{P_n} &= \sum_{i=1}^{n} f(c_i)\Delta x_i \\ &= \sum_{i=1}^{n} \left(\frac{i}{n} + \frac{i^2}{n^2}\right)\frac{1}{n} \\ &= \frac{1}{n^2}\sum_{i=1}^{n} i + \frac{1}{n^3}\sum_{i=1}^{n} i^2 \\ &= \frac{1}{n^2}\frac{n(n+1)}{2} + \frac{1}{n^3}\frac{n(n+1)(2n+1)}{6} \\ &= \frac{1}{2}\left(1+\frac{1}{n}\right) + \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right). \end{aligned}$$

As $\lim_{n\longrightarrow\infty} S_{P_n} = \frac{5}{6}$, we have

$$\int_0^1 \left(x + x^2\right) dx = \frac{5}{6}.$$

Homework

For the following continuous functions find a formula for the Riemann sum obtained by dividing the interval $[a, b]$ into n equal subintervals and using the right-hand endpoint for each c_i. Then take a limit of these sums as $n \to \infty$ to compute the corresponding Riemnann integral.

1. $f(x) = x^2 + 1$ over the interval $[0, 3]$.
2. $f(x) = x^2 - x^3$ over the interval $[-1, 0]$.
3. $f(x) = 2x^3$ over the interval $[0, 1]$.

Properties of Riemann Integration

Theorem

Let f and g be Riemann integrable over the interval $[a, b]$. Then

(1) *Order of Integration:* $\int_b^a f(x)dx = -\int_a^b f(x)dx$ *(definition)*

(2) *Zero Width Interval:* $\int_a^a f(x)dx = 0$ *(a definition when $f(a)$ exists)*

(3) *Constant Multiple:* $\int_a^b kf(x)dx = k\int_a^b f(x)dx$

(4) *Sum:* $\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)dx + \int_a^b g(x)dx$

Difference: $\int_a^b (f(x) - g(x))\,dx = \int_a^b f(x)dx - \int_a^b g(x)dx$

Theorem *Contd.*

(5) Additivity: $\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$

(6) Max-Min Inequality: If f has maximum value M and minimum value m on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$$

(7) Domination: If $f(x) \leq g(x)$ on $[a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.

Theorem (The Fundamental Theorem of Calculus: part 1)

If f is Riemann integrable on $[a, b]$ and if there is a differentiable function F on $[a, b]$ such that $F' = f$, then

$$\int_a^b f(x)dx = F(b) - F(a).$$

Proof:

Let $\epsilon > 0$ be a fixed positive number. As f is Riemann integrable, corresponding to the given ϵ, there exist a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that $U(P, f) - L(P, f) < \epsilon$.

Given that the function F is differentiable on $[a, b]$, so it both differentiable and continuous in every subinterval $[x_{i-1}, x_i]$. Thus, by the mean value theorem there exists a t_i in $[x_{i-1}, x_i]$ such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(t_i) \qquad \text{for } i = 1, 2, \ldots, n.$$

$$\implies \frac{F(x_i) - F(x_{i-1})}{\Delta x_i} = f(t_i) \qquad \text{for } i = 1, 2, \ldots, n.$$

$$\implies F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i \qquad \text{for } i = 1, 2, \ldots, n.$$

Thus

$$\sum_{i=1}^{n} f(t_i)\Delta x_i = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = F(b) - F(a).$$

Proof continued

As

$$L(P,f) \leq \sum_{i=1}^{n} f(t_i)\Delta x_i \leq U(P,f) \quad \text{and} \quad L(P,f) \leq \int_a^b f(x)dx \leq U(P,f)$$

and

$$U(P,f) - L(P,f) < \epsilon,$$

we have

$$\left| \sum_{i=1}^{n} f(t_i)\Delta x_i - \int_a^b f(x)dx \right| < \epsilon$$

$$\implies \left| F(b) - F(a) - \int_a^b f(x)dx \right| < \epsilon.$$

Since this holds for every ϵ, $\int_a^b f(x)dx = F(b) - F(a)$, and the theorem follows.

Theorem (The Mean Value Theorem for Definite Integrals)

Let $f : [a, b] \longrightarrow \mathbb{R}$ be a continuous function. Then there exists $c \in [a, b]$ such that

$$\frac{1}{b-a}\int_a^b f(x)dx = f(c).$$

Proof:

As every continuous function $f(x)$ defined over a closed bounded interval $[a, b]$ attains its maximum M and minimum m, we have

$$m \leq f(x) \leq M \qquad \text{for all } x \in [a, b]$$

$$\implies \int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx$$

$$\implies m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$\implies m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M.$$

As f is continuous on $[a, b]$, the intermediate value theorem implies that there exists $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

Theorem (The Fundamental Theorem of Calculus: part 2)

Let f be Riemann integrable on $[a, b]$. For $a \leq x \leq b$, put

$$F(x) = \int_a^x f(t)dt.$$

Then F is continuous on $[a, b]$. Further, if f is continuous on $[a, b]$, then F is differentiable on $[a, b]$ and $F' = f$.

Proof:

Let c be a point in $[a, b]$. To show that F is continuous at c, we shall show that, for every $\epsilon > 0$, there exists a $\delta > 0$ such that, if $|x - c| < \delta$, then $|F(x) - F(c)| < \epsilon$.

Since f is Riemann integrable, it is bounded. That is $|f(x)| \leq M$ for some $M \geq 0$. It follows that

$$|F(x) - F(c)| = \left| \int_a^x f(t)dt - \int_a^c f(t)dt \right| = \left| \int_c^x f(t)dt \right| \leq M \left| \int_c^x dt \right| = M|x - c|.$$

Let $\epsilon > 0$ be a positive number. Choose $\delta = \dfrac{\epsilon}{M}$. Then, if $|x - c| < \delta$, then $|F(x) - F(c)| < \epsilon$. This implies F is continuous at c.

Proof Continued: By the definition of derivative

$$\begin{aligned} F'(x) &= \lim_{h\to 0} \frac{F(x+h)-F(x)}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left[\int_a^{x+h} f(t)dt - \int_a^{x} f(t)dt\right] \\ &= \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)dt. \end{aligned}$$

According to the Mean Value Theorem for definite integrals, there is some point c between x and $x+h$ such that

$$f(c) = \frac{1}{h}\int_x^{x+h} f(t)dt.$$

Thus,

$$F'(x) = \lim_{h\to 0} f(c).$$

Further, as $h \to 0$, $x+h$ approaches x, which forces c to approach x. Since f is continuous at x, $f(c)$ therefore approaches $f(x)$. This shows that F is differentiable at x and $F'(x) = f(x)$.